Explainable AI for medical imaging: deep-learning CNN ensemble for classification of estrogen receptor status from breast MRI

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ABSTRACT

Deep-learning convolutional neural networks (DCNNs) are the most commonly used approach in medical image analysis tasks at present; however, they have largely been used as black-box predictors, lacking explanation for the underlying reasons. Explainable artificial intelligence (XAI) is an emerging subfield of AI seeking to understand how models make their decisions. In this work, we applied XAI visualization to gain an insight into the features learned by a DCNN trained to classify estrogen receptor status (ER+ vs ER-) based on dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) of the breast. Our data set contained 1395 ER+ regions-of-interest (ROIs) and 729 ER- ROIs from 148 patients, each with a pre-contrast scan and a minimum of two post-contrast scans. We developed a novel transfer-trained dual-domain DCNN architecture derived from the AlexNet model trained on ImageNet data that received the spatial (across the volume) and dynamic (across the acquisition sequence) components of each DCE-MRI ROI as input. The network's performance was evaluated with the area under the receiver operating characteristic curve (AUC) from leave-one-case-out cross-validation. To visualize the DCNN learning, we applied XAI techniques, including the Integrated Gradients attribution method and the SmoothGrad noise reduction algorithm, to the ROIs from the training set. We observed that our DCNN learned relevant features from the spatial and dynamic domains, but there were differences in the contributing features from the two domains. We also visualized DCNN learning from irrelevant features resulting from pre-processing artifacts. These observations motivate new approaches to pre-processing our data and training our DCNN.

Keywords: explainable artificial intelligence, deep learning, medical imaging, interpretable AI, breast cancer, magnetic resonance imaging, estrogen receptor, transfer learning

1. INTRODUCTION

Using deep-learning predictive models as black-box classifiers is problematic, especially in the medical domain because models can exploit irrelevant patterns in the data to maximize the performance metrics.[1] This pitfall of black-box deep learning models demonstrates Campbell's law of economics[2] cautioning that societal systems evaluated by well-meaning metrics are susceptible to corruption by pressures to optimize them. Across the artificial intelligence (AI) community, there are a plethora of anecdotes of AI models[3] learning unintended and creative solutions to tasks that rely on a performance metric guiding the model's learning. With Campbell's law influencing AI systems, it is imperative that AI users are aware of the underlying reasons that contribute to the decision made by the deep-learning models. Using explainable AI (XAI)[4] algorithms, we generated attribution maps for breast dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) data highlighting features contributing to the output score. This novel XAI method expands on previous techniques for explaining model predictions by using methods applicable to any deep-learning convolutional neural network (DCNN) models, and accounting for cross-correlation between input features and diverging from other perturbation-based methods (i.e. occlusion) that only consider the effect of individual features on the prediction[5]. This analysis may be used to examine if the predictive model is guided by relevant features during training, and help develop custom strategies to optimally train a DCNN.

In this study, we trained a DCNN for the classification of estrogen receptor status (ER+ and ER-) from DCE-MRI breast volumes to aid in the molecular classification of breast cancer. Three types of analysis were deduced from the generated attribution maps: the DCNN learning from (a) expected relevant features, (b) complementary patterns between spatial and dynamic MRI components, and (c) irrelevant features produced from preprocessing steps. In current clinical practice, immunohistochemistry test of biopsy samples is used to evaluate ER status. The long-term goal of our work is to develop non-invasive surrogate biomarkers that have the potential to replace invasive procedures for breast cancer treatment planning.

Medical Imaging 2020: Computer-Aided Diagnosis, edited by Horst K. Hahn,
Maciej A. Mazurowski, Proc. of SPIE Vol. 11314, 113140Z · © 2020 SPIE
CCC code: 1605-7422/20/$21 · doi: 10.1117/12.2549298

2. MATERIALS AND METHODS

2.1 Data sets

The data set consisted of T1-weighted DCE-MRI scans from six institutions with a total of 148 patients diagnosed with stage 2 and 3 invasive ductal carcinoma undergoing neoadjuvant chemotherapy[6]. The ER status of each breast cancer was provided with the data set. An experienced breast radiologist marked a 3-dimensional (3D) bounding box enclosing the tumor for each DCE-MRI scan, amounting to 2,124 slice images to train and test the DCNN. Each MRI volume was down-sampled to an in-plane size of 128x128 pixels.

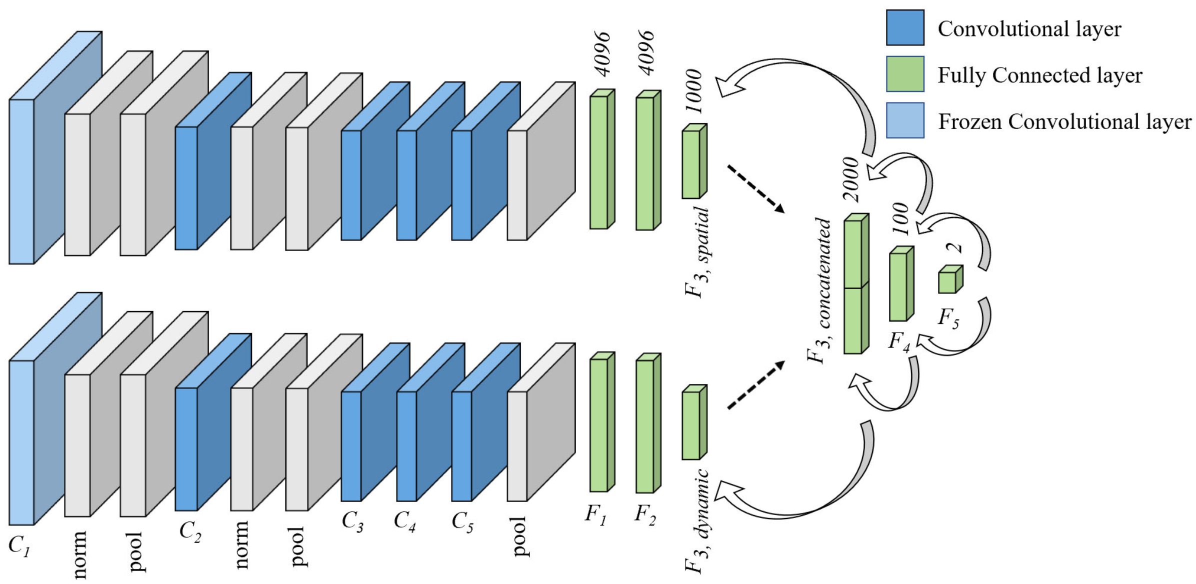


Fig. 1. Dual domain DCNN structure for breast cancer subtype classification, with identical structures adapted from AlexNet. The spatial ROI component was inputted into the top domain and the dynamic ROI component was inputted into the bottom domain. The arrows demonstrate the backpropagation that updates the weights of both domains simultaneously.

2.2 Spatial and dynamic domains

For the spatial region-of-interest (ROI) domain, we first excluded the pectoral muscle and background external to the breast to reduce the effects on the homogenization process. The process involved applying Whitestripe normalization[7] on the DCE-MRI volumes at peak contrast to homogenize the data using the non-dense region of adipose tissue as a reference. This technique had been shown to enable more generalizable training by reducing imaging center dependency.[8] For each image slice within the radiologist provided bounding box of the volume, one slice above and one slice below were extracted. These ROIs were then stored as a 3-channel image in the order they appeared in the volume. For the dynamic ROI domain, we also excluded the pectoral muscle and background noise to ensure the spatial and dynamic images contained the same regions of the breast. We then extracted the ROI from three time points of the original MRI scans without homogenization and stored these ROIs as another 3-channel image in the order of the acquisition sequence. Note that the lesion was not always centered in the ROI.

2.3 Dual-domain DCNN structure and transfer learning

We developed a novel dual-domain DCNN architecture to utilize the spatial and dynamic components of DCE-MRI volumes for classifying ER status. Our model consisted of two identical DCNNs creating a two-input mode mirroring the design of AlexNet that won the ImageNet LSVRC-2012 contest.[9] The structure of each domain in this DCNN ensemble (fig.1) has five convolutional layers and three fully connected layers with a total of 120M parameters for the entire ensemble. The last fully connected layer of each domain was concatenated and processed by two additional fully connected layers (F_4 and F_5) that reduced the original feature space of 2000 nodes to two output classes. For the DCNN training, the 3-channel spatial images of a given ROI into the designated spatial domain path and the 3-channel dynamic images of the ROI into the dynamic domain path were inputted simultaneously. Thus, the weights of both domains were updated simultaneously after each iteration using the same output score during backpropagation, making the spatial and dynamic attribution maps complementary (see Section 2.4). We denote the n convolutional layers as $C_1, \ldots, C_n,$ where n=5 for AlexNet. For transfer learning, both domains were initialized with the weights transferred from training on the ImageNet data. The weights of C_1 for both paths was frozen, which we found to be a good strategy when the target domain training set was small.[10] The DCNN was trained and evaluated using leave-one-case-out cross-validation.

2.4 Explainable AI

Attribution maps (fig. 2) were generated using the backpropagation-based integrated gradients attribution method[11] and using the Smoothgrad algorithm[12] to reduce the noise in the map. An *attribution method* accredits input features for the output score of the DCNN by assigning more influential pixels with higher values, thus connecting the complicated path from the predicted output score to the patterns observed at every layer of the DCNN. The integrated gradients method attributes the influence of each pixel by considering the input image relative to an uninformative (i.e. score is approximately equal to zero) baseline image, which, in our case, was chosen to be an image with all pixel values set to zero. The integrated gradients attribution map is defined as the integral of the gradients, with respect to the output score, along the straight line in $\mathbb{R}^n$ from the baseline to the input image, where n is the number of pixels in the input image. This integral is calculated by Riemann approximation, summing the gradients at evenly spaced points along the path in $\mathbb{R}^n$.

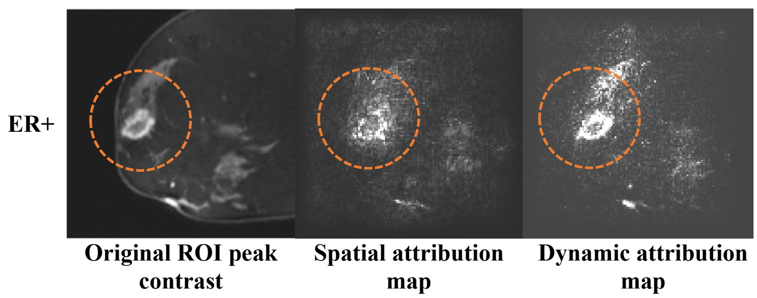


Fig. 2. Original ROI at peak contrast (*left*) with its corresponding spatial (*middle*) and dynamic (*right*) attribution maps. For interpreting an attribution map, note that the higher the brightness of a pixel, the more relevant they are to the model's prediction. In this case, we can see a high density of bright pixels corresponding to the tumoral and peritumoral tissue marked by the circle.

3. RESULTS

The DCNN ensemble in this study was trained on the classification task of ER status from DCE-MRI breast volumes across 6 imaging centers. The AUC recorded from the leave-one-case-out cross-validation was 0.67. The attribution maps generated for this study were obtained by deploying the trained DCNN on the training data set. Thus, these cases were correctly classified and received correct prediction scores >0.9. By visualizing the attribution maps for the training set, we can gain a better insight into the input features learned by the DCNN.

3.1 Learning relevant features

In this section, we discuss the expected relevant features learned by the DCNN. The example in fig. 3 shows that the DCNN learned most of the features from the ROI at and after peak contrast. This observation was largely expected because the contrast accentuated the blood vessels and tumor. We also found that for many cases (fig. 4) the network distinguished the tumor from surrounding fatty, dense, and scattered tissue for both the spatial and dynamic ROI components. This suggests that the network considered the tumoral and peritumoral tissue to be relevant features for its prediction.

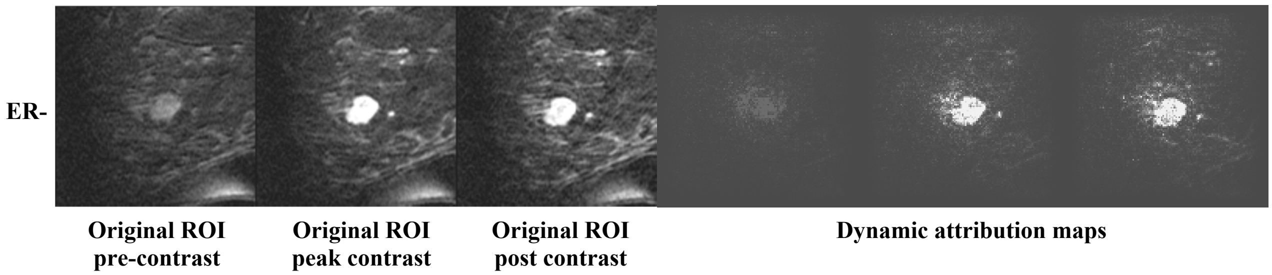


Fig. 3. Attribution maps from dynamic ROIs, demonstrating the network learning the most from the central ROI slice at peak.

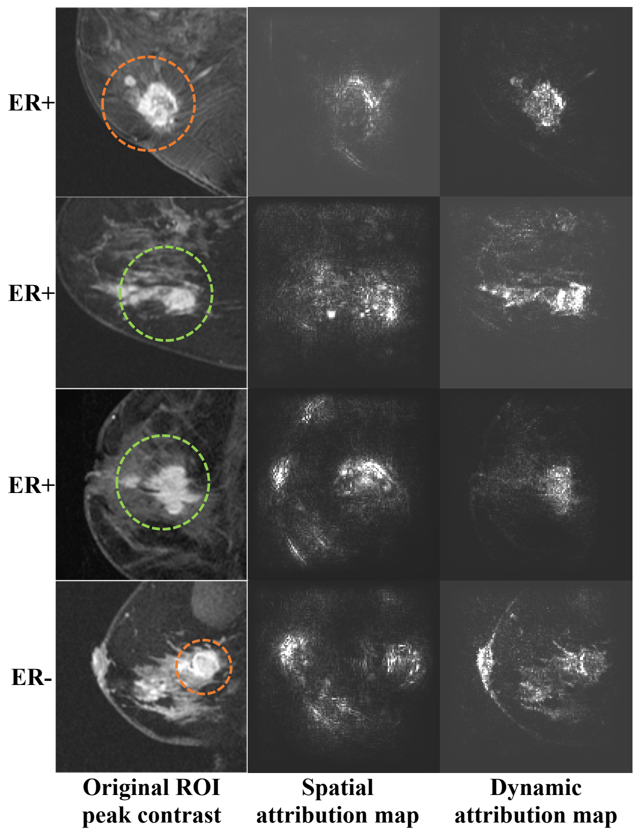


Fig. 4. Spatial and dynamic attribution maps for ROIs at peak contrast demonstrating the network learning from relevant features, circled in ROI slice, for both components.

3.2 Differences in learning between spatial and dynamic domains

In this section, we discuss notable differences in learning from the spatial and dynamic DCE-MRI domains. First, we observe that the DCNN learned from the fatty tissue surrounding the tumor in the spatial domain more frequently than the dynamic domain (fig. 5). We hypothesize this is a result of the breast structure changing between the 3 channels of the spatial ROI component, which may have caused the network to learn the textural features of the input ROI. Second, we observe that the DCNN distinguishes the tumoral tissue from fatty and dense tissue in the dynamic ROI more effectively than the spatial ROI (fig. 6). We hypothesize that this is a result of the pre-contrast image preceding the peak contrast image in the 3-channel ROI; the image of the breast without contrast may have served as a reference for the network to learn from the various tissues and blood vessels that the contrast agent accentuated in the breast image after contrast arrived. These observations show that each component presents new information to the network, reinforcing the need to learn both components.

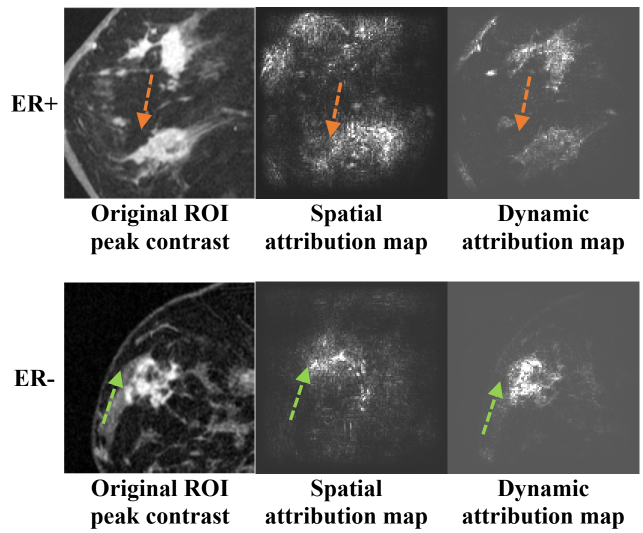
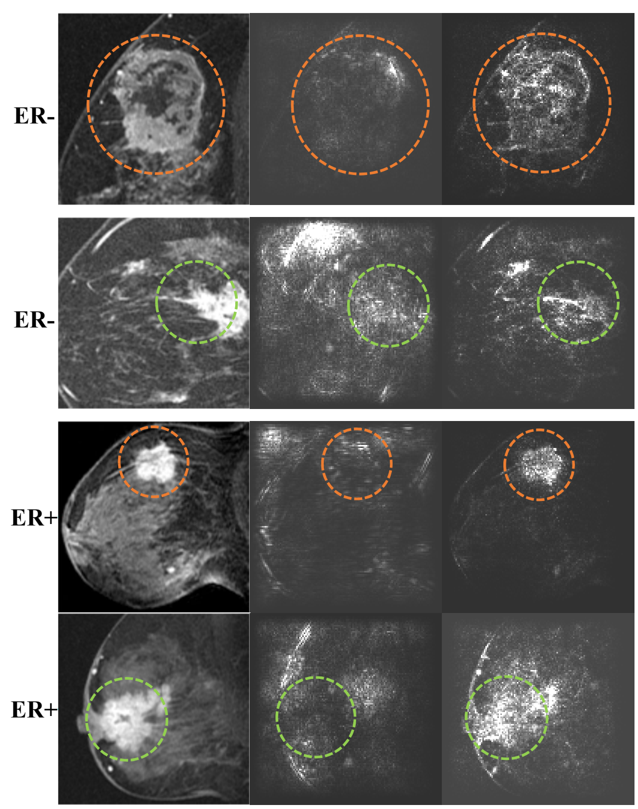


Fig. 5. Instances of spatial attribution maps having higher densities of bright pixels in tissue near the tumor, showing that the DCNN learned features from the spatial but not in the dynamic component.



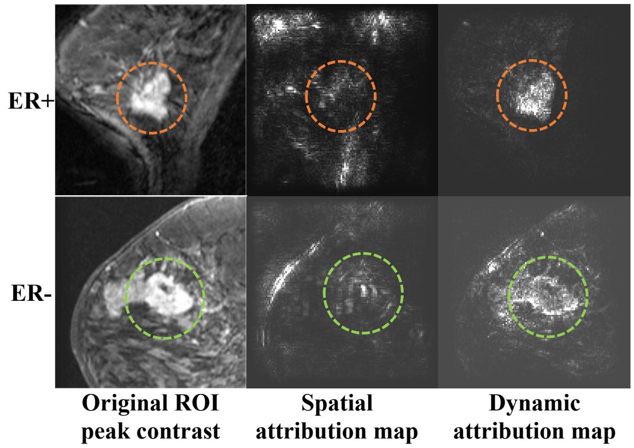


Fig. 6. Instances of dynamic attribution maps having higher densities of bright pixels on tumoral tissue of the ROI at peak contrast than the spatial attribution maps, suggesting that the DCNN learned more tumoral features from the dynamic MRI component.

3.3 Learning from irrelevant features from pre-processing

In this section, we discuss a possible flaw in the pre-processing of the ROIs that may have misguided the network's training. In a few of the ROI attribution maps there were high densities of bright pixels along edges of the breast (fig. 7). We hypothesize that these regions were mostly formed as a result of cropping the pectoral muscle and background noise for homogenization. We can also see, most prominently in the upper two cases of figure 7, that the high densities of bright pixels are near the thickened skin of the breast. Possibly excluding these features of the breast from the training could force the model to learn from the tissue and blood flow of the breast. These edges contributing to learning of the network may cause sub-optimal or misleading results. By eliminating these irrelevant features, it is possible to improve the DCNN's ability to generalize to the test set.

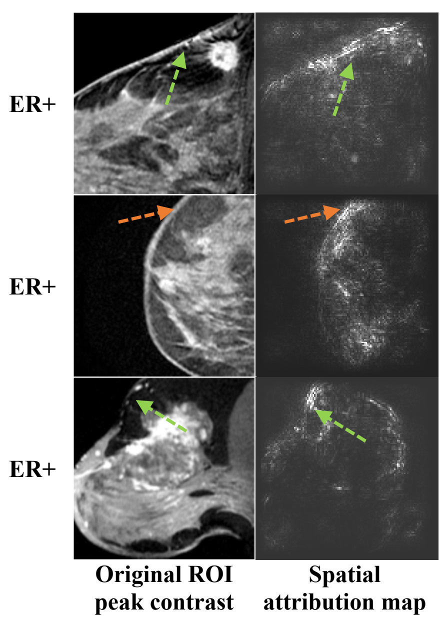


Fig. 7. Examples of the spatial attribution maps affected by edges generated from pre-processing.

4. DISCUSSION AND CONCLUSION

There are two key motivations behind applying explainable AI to deep learning on our medical imaging data set. First, because our data set is small, we can use these attribution maps to identify artifacts that may interfere with the learning. The information may provide guidance to improve our preprocessing steps and fine-tune our model to learn relevant features from the DCE-MRI breast ROIs. Second, we can gain better insight into the input features that contribute to the output prediction of the model, presenting an opportunity to learn imaging characteristics that may distinguish between ER+ and ER- patient cases. We also presented a novel dual-domain DCNN that can be generalized to multi-domain and/or multi-modality for other medical image analysis tasks. This model architecture allows the DCNN to select features from various complementary characteristics of images of the same patient.

The future direction of this work should aim to remove the preprocessing artifacts of the breast ROIs. The model should then be re-trained and deployed to determine if the performance improves and attribution maps should also be generated to determine if the model is learning from relevant features. More work should also be done to detect any differences in learning between ER- vs. ER+ and spatial vs. dynamic DCE-MRI breast ROIs. Also, we could construct our dual domain architectures using other architectures. Finally, increasing our data set by recovering more labels from The Cancer Imaging Archive or introducing a third class of DCE-MRI without cancer could help our model distinguish between the ER+ and ER- better.

This work focuses on understanding what the DCNN learns behind its output decision because studies have shown that AI could learn unintended solutions to problems by exploiting irrelevant patterns in data sets. The field of medical imaging is more susceptible to this issue because of the lack of massive training set. This pilot study intends to take a step toward understanding the features that these complex networks with large capacity may have learned, which may be useful for guiding the development of methods to minimize corruption in their learning and, in the long run, may improve the interpretability of DCNN and advance AI toward intelligent decision support tools in medicine.

ACKNOWLEDGMENTS

This work is supported in part by National Institutes of Health award number R01 CA214981. RKS is also supported by the Basic Radiological Science New Investigator Award from the Department of Radiology at the University of Michigan.

REFERENCES

[1] Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., and Oermann, E. K., "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study," PLoS medicine*,* 15(11), (2018).

[2] Campbell, D. T., "Assessing the impact of planned social change," Occasional Paper Series*,* 8, (1976).

[3] Lehman, J., Clune, J., and Misevic, D., "The surprising creativity of digital evolution," Artificial Life Conference Proceedings, 55-56 (2018).

[4] Adadi, A., and Berrada, M., "Peeking inside the black-box: A survey on Explainable Artificial Intelligence (XAI)," IEEE Access*,* 6, 52138-52160 (2018).

[5] Ancona, M., Ceolini, E., Öztireli, C., and Gross, M., "A unified view of gradient-based attribution methods for deep neural networks," NIPS 2017-Workshop on Interpreting, Explaining and Visualizing Deep Learning, (2017).

[6] Lingle, W., Erickson, B., Zuley, M., Jarosz, R., Bonaccio, E., Filippini, J., and Gruszauskas, N., "Radiology Data from The Cancer Genome Atlas Breast Invasive Carcinoma [TCGA-BRCA] collection," The Cancer Imaging Archive, (2016).

[7] Shinohara, R. T., Sweeney, E. M., Goldsmith, J., Shiee, N., Mateen, F. J., Calabresi, P. A., Jarso, S., Pham, D. L., Reich, D. S., and Crainiceanu, C. M., "Statistical normalization techniques for magnetic resonance imaging," NeuroImage: Clinical*,* 6, 9-19 (2014).

[8] Samala, R. K., Chan, H.-P., Hadjiiski, L., Paramagul, C., Helvie, M. A., and Neal, C. H., "Homogenization of breast MRI across imaging centers and feature analysis using unsupervised deep embedding," Proc. SPIE*,* 10950, 10950201-7 (2019).

[9] Krizhevsky, A., Sutskever, I., and Hinton, G. E., "Imagenet classification with deep convolutional neural networks," Advances in Neural Information Processing Systems, 1097-1105 (2012).

[10] Samala, R. K., Chan, H.-P., Hadjiiski, L. M., Helvie, M. A., Richter, C. D., and Cha, K., "Breast Cancer Diagnosis in Digital Breast Tomosynthesis: Effects of Training Sample Size on Multi-Stage Transfer Learning using Deep Neural Nets," IEEE Transactions on Medical Imaging, 38(3), 686-696 (2019).

[11] Sundararajan, M., Taly, A., and Yan, Q., "Axiomatic attribution for deep networks," Proceedings of the 34th International Conference on Machine Learning-Volume 70, 3319-3328 (2017).

[12] Smilkov, D., Thorat, N., Kim, B., Viégas, F., and Wattenberg, M., "Smoothgrad: removing noise by adding noise," arXiv preprint arXiv:1706.03825, (2017).